04 Organic Structures Problem-solving Approach

Basis: from the 2016–2025 Chemistry ATAR past exam questions in Section Two and Section Three that are labelled ORG, 9 patterns in total, 98 label occurrences in total (consistent with INDEX.md; one part may carry multiple labels; excludes Section One multiple-choice questions). Marking points are taken from the marking criteria key, paraphrased, not copied verbatim; the original papers and marking files remain in a local ignored directory. Exam-question numbering format: `2024S-Q35(a)` = 2024 Section Two Question 35(a); `2019E-Q37(e)` = 2019 Section Three Question 37(e). A whole question with no sub-questions is written as `2018S-Q34`. Pattern definitions are taken from `tools/catalogue_CHE.json`. This file has undergone independent review (against past exam questions, keys and exam reports); records are in `CHE/methods/审校记录.md`. ★ marker: arranged by total marks for each pattern (INDEX.md), taking approximately the top quarter; this file covers ORG4 and ORG2.

Marking panel reminders

- Solubility and boiling-point explanations are the hardest category: 2021 report: the solubility question (2021S-Q33, solubility of ethanol and decan-1-ol in water) was the hardest question on the paper, average 30.7%. 2024 report: 2024S-Q28 (ibuprofen solubility) was one of the hardest questions. 2025 report: 2025E-Q37(c) (difference in solubility of nonanoic acid and methanol in water) was the hardest question.
- Structural formulae and naming are relatively stable: 2021 report: questions on identifying structural formulae and naming errors averaged 77.5%. 2025 report: questions on drawing structural formulae and naming were answered relatively well (2025S-Q31).
- Long questions are often left blank: 2020 report: 2020S-Q33 (synthesis of ethyl pentanoate, involving ORG5, ORG6, ORG7) averaged 39.53%, with many candidates leaving it blank.
- key requirements for "explain": A complete explanation must include both "which force" and the "energy balance" (whether the new forces formed can overcome the old forces broken); merely naming the force does not earn full marks.

Pattern 1: Identifying Functional Groups and Classes of Organic Compounds

Identify: the question gives a structural formula and asks to "circle and name", "identify the functional group", "state the class of compound"; or gives a compound and asks which class it belongs to (soap, ester, alkene, etc.).

Steps

1. Scan the structure one by one for characteristic fragments: C=C (alkene), C–O–H (alcohol), C=O attached to H (aldehyde), C=O within the chain (ketone), –COOH (carboxylic acid), –COO– (ester), –NH_2 (amine), –CONH– (amide).
2. Each functional group corresponds to one name; when writing names, use the standard term (hydroxyl / alcohol, carboxyl / carboxylic acid, amino / amine are all accepted).
3. When circling, circle the whole functional group (e.g. for an ester, circle the entire –COO– fragment; for an amide, include C=O and N).
4. When acidic and basic groups are present together in a structure (e.g. cysteic acid, which contains both a sulfonic acid group and an amino group, questions like 2017E-Q40(f)), label them separately as acidic or basic.
5. When asked for the "class", write the class name rather than the structural formula: e.g. for the class of soap, write soap / salt / carboxylate.

Marking points (typical wording in the key): 1 mark for each correctly circled and named functional group, capped at the question's marks (2016S-Q36(c): 1 mark per functional group, 3 marks total); the class name is consistent with the structure (2020E-Q40(eii) accepts soap, salt, carboxylate); the acidic or basic label is correct (2017E-Q40(f)).

Common errors (exam reports)

- Confusing aldehydes and ketones: an aldehyde's C=O is at the end of the chain and attached to H; a ketone's C=O is within the chain. The key for 2025E-Q38(a) requires the correct names and structures of a carboxylic acid and an aldehyde/ketone.
- A soap structure requires both the carboxylate head and the long hydrocarbon chain (2017E-Q39(cii)); writing only "fatty acid" does not earn full marks.
- The 2021 report states that questions on identifying structural formulae and naming errors were answered relatively well (average 77.5%); this pattern is a relatively stable source of marks in ORG.

Past exam questions: 2016S-Q36(c), 2017E-Q39(ci), 2017E-Q39(cii), 2017E-Q40(f), 2018S-Q34, 2019E-Q36(h), 2020E-Q40(eii), 2020S-Q28(bi), 2021E-Q34(d), 2021S-Q26(a), 2025E-Q38(a)

Pattern 2: IUPAC Naming and Structural Formula Interconversion ★

Identify: the question asks for an "IUPAC name", "draw the structure showing all atoms and bonds", "complete the table"; or gives the number of carbons, functional group and isomer type, and asks you to draw the structural formula.

Steps

Naming from a structural formula

1. Find the longest carbon chain containing the principal functional group as the parent (meth–, eth–, prop–, but–, pent–, hex–, hept–, oct–).
2. Number so that the principal functional group has the lowest locant; when the principal functional group locants are the same, make the C=C and substituent locants as low as possible.
3. When writing substituents (alkyl), arrange them alphabetically, put the locant before each, and use di-, tri- for identical substituents.
4. Suffixes: -ol (alcohol), -one (ketone), -al (aldehyde), -oic acid (carboxylic acid), -amine (amine), -amide (amide), -ene (alkene).
5. Esters are named as "alkyl + alkanoate", with the alkyl group first (e.g. ethyl pentanoate).

Drawing a structural formula from a name

1. First draw the parent chain, then place the principal functional group on the specified carbon according to the locant.
2. Complete the hydrogens on each carbon so that each carbon has 4 bonds; double bonds and C=O must be drawn explicitly.
3. Check: the number of carbon atoms matches the name; all H atoms and bonds have been drawn (questions often say "show all hydrogen atoms").

Marking points (typical wording in the key): a completely correct structural formula earns full marks; only one small error or omission (e.g. one missing H) usually earns only 1 mark (the keys for 2020S-Q26 and 2016S-Q28(a) both have this rule); the name is consistent with the drawn structure (2016S-Q29(a), (b)); the "correct IUPAC name" mark is awarded only if both the name and structure are correct (2021S-Q26(a)). The key also accepts older-style names such as 2,3-dimethyl-2-butene.

Common errors (exam reports)

- Numbering does not take the lowest locants. Students' pentan-4-one in 2021S-Q26(a) is an example; the correct parent numbering should start from the other end, giving pentan-2-one.
- Missing H atoms or missing bonds. 2025S-Q31 requires "Show all hydrogen atoms and all bonds"; omissions lose marks. The 2025 report states that structural-formula drawing and naming questions were answered relatively well, but each item still needs checking.
- The key for 2019S-Q33 (12 marks) can be checked directly: the product of pent-2-ene and Br_2(aq) is 2,3-dibromopentane (1 mark each for the chain and Br positions, 1 mark for the name); ethanal is oxidised by acidified $KMnO_4$ to ethanoic acid (structure and name of ethanoic acid); butanoic acid reacts with Na_2CO_3 to form butanoate (sodium salt). For each row, the reactant structure, product structure and name must all be written in full.

- 2019E-Q37(c): the soap ion $C_{17}H_{31}CO_2^-$ must be drawn with all bonds, including the two C=C bonds, and must contain COO^-.

Past exam questions: 2016S-Q28(a), 2016S-Q29(a), 2016S-Q29(b), 2016S-Q35, 2019E-Q36(h), 2019E-Q37(c), 2019E-Q38(b), 2019S-Q33, 2020S-Q26, 2021S-Q26(a), 2021S-Q26(b), 2022S-Q27(a), 2022S-Q32(a), 2022S-Q33(b), 2023E-Q37(c), 2023E-Q37(fii), 2024S-Q35(a), 2025S-Q31

Pattern 3: Isomers

Identify: the question gives a molecular formula or a structure and asks to "draw a different isomer", "draw two isomers", "draw the three isomers" or "cis/trans"; or asks you to select isomers according to chemical properties (whether they can be oxidised).

Steps

1. Calculate the degree of unsaturation from the molecular formula and determine the possible classes (alkene/cycloalkane, alcohol/ether, aldehyde/ketone, carboxylic acid/ester).
2. List by isomer type: chain isomers, positional isomers (double-bond or functional-group position), functional-group isomers (e.g. aldehyde and ketone, carboxylic acid and ester), cis/trans isomers.
3. Cis/trans isomerism exists only when both ends of the double bond have two different groups attached. In a 1-ene (e.g. but-1-ene), C1 is attached to two H atoms, so there is no cis/trans isomerism.
4. For each structure, draw all atoms and bonds and give the correct name.
5. If the question requires distinguishing properties (e.g. "can be oxidised by acidified potassium dichromate"), decide by functional group: primary alcohols and aldehydes can be oxidised; ketones cannot.

Marking points (typical wording in the key): 2 marks for each correct structure, 1 mark if there is one small error (e.g. one missing H) (wording in the keys for 2016S-Q36(d) and 2025S-Q34); the isomer type required by the question must match (2017S-Q35(a): one cis and one trans); the property must match (2018S-Q34: one oxidisable aldehyde, one non-oxidisable ketone).

Common errors (exam reports)

- Confusing the oxidisability of aldehydes and ketones: the key for 2018S-Q34 states clearly that an aldehyde can be oxidised, while a ketone (e.g. 2-hexanone) cannot.
- Writing a 1-alkene as having cis/trans isomerism: the key for 2023E-Q37(c) gives but-1-ene, cis-/trans-but-2-ene (the C_5H_{10} version is pent-1-ene, cis-/trans-pent-2-ene); cis/trans isomers occur only in 2-enes.
- Confusing acids and esters: 2025S-Q34 ($C_3H_6O_2$) requires one carboxylic acid and two esters; all three structures must be correct, and for esters the two sides of "R–COO–R′" must be distinguished.
- Reports: the 2023 report states that Question 37 was generally answered well (the report does not comment separately on (c)). The 2025 report states that 2025S-Q34 (drawing three isomers, showing all atoms and bonds) was one of the best-answered questions in Section Two.

Past exam questions: 2016S-Q36(d), 2017S-Q35(a), 2018S-Q34, 2023E-Q37(c), 2025S-Q34

Pattern 4: Intermolecular forces explaining boiling point and solubility ★

Identify: The question may ask "account for the difference in boiling points", "explain the solubility", "explain why … dissolves/does not dissolve", "using your understanding of intermolecular forces"; or give a table of boiling points or solubilities and require an explanation.

Steps

1. List the intermolecular forces for each substance: all molecules have dispersion forces; molecules with a permanent dipole (e.g. C=O, C–Cl) have dipole–dipole forces; molecules with H attached to F, O or N in O–H or N–H have hydrogen bonding; between ionic compounds and polar molecules there are ion–dipole forces.

2. Boiling point: boiling requires overcoming intermolecular forces. First compare the strongest force (hydrogen bonding > dipole–dipole > dispersion forces), then, when the forces are the same, compare dispersion forces (the larger the molecule and the longer the carbon chain, the stronger the dispersion forces).
3. Solubility: whether solute and solvent are miscible depends on whether the new forces formed can release enough energy to replace the original solute–solute and solvent–solvent interactions. Write the three sets of interactions (solute–solute, solvent–solvent, solute–solvent).
4. Conclusion: state "the forces formed are sufficient to overcome the forces broken, so it dissolves" or "they are not sufficient, so solubility is low".
5. When drawing diagrams: show hydrogen bonds with dashed lines, and label the type of force in the diagram (labelled diagram).

Mark-earning points (typical wording in the key): correctly name each type of force and match the force to the substance (2016S-Q37: pentane has only dispersion forces, pentanal adds dipole–dipole, pentanoic acid adds hydrogen bonding, 1 mark each); write the energy balance (2017S-Q32(a): hydrogen bonds between water molecules and between ethanol molecules are broken, and new hydrogen bonds are formed); the explanation must include both "overcoming" and "forming" (the 2021S-Q33 key allows either an "energy response" or a "strength response").

Common errors (examiner report)

- Only saying "like dissolves like" or "similar polarity", without writing the specific forces and energy. 2017S-Q32(b) requires explaining why the solubility of alcohols in water decreases as the hydrocarbon chain grows; the key is that dispersion forces increase with chain length, and the new forces formed are not sufficient to overcome the original forces.
- Explaining the high melting point of ionic crystals or zwitterions as "breaking covalent bonds" (2023E-Q36(c), 2024S-Q29(b)); the key requires writing that ionic bonds (electrostatic attraction) need more energy to overcome.
- Difficult questions named in the report: 2022S-Q30 (why milk removes the burning sensation of capsaicin better than water, requiring comparison of the forces of fat and water); 2023S-Q32 (reason for dichloromethane as a solvent for caffeine); 2024S-Q28 (ibuprofen); 2025E-Q37(c) (nonanoic acid and methanol).
- The 2019E-Q41(d) key requires explaining that proteins do not dissolve in water: the dispersion forces between the large molecules are very large, and the forces formed by water molecules are not sufficient to break them.

Past questions: 2016S-Q35, 2016S-Q37, 2017E-Q39(aii), 2017E-Q39(aiii), 2017E-Q40(d), 2017S-Q32(a), 2017S-Q32(b), 2017S-Q32(c), 2018S-Q33(b), 2019E-Q37(a), 2019E-Q41(d), 2020S-Q32(e), 2021S-Q33, 2022S-Q27(d), 2022S-Q30, 2023E-Q36(c), 2023S-Q32, 2024S-Q28, 2024S-Q29(b), 2025E-Q35(b), 2025E-Q37(c)

Pattern 5: Addition reactions

Identify: The question gives an alkene and asks for the product of adding Br_2, H_2, HX or water (hydration); or asks for an equation; or uses the colour change of bromine water to test for unsaturation.

Steps

1. Find the C=C and determine the reagent added.
2. Break the π bond, with one atom or group adding to each carbon: Br_2 gives a 1,2-dibromoalkane; hydration (H^+ catalysed) gives an alcohol, with the major product determined by Markovnikov's rule.
3. Write the product name and structural formula (all H and bonds).
4. For equations use condensed structures, write the catalyst and conditions, and balance.
5. Test: bromine water changing rapidly from orange to colourless indicates an unsaturated bond; a cycloalkane does not decolourise or does so very slowly.

Mark-earning points (typical wording in the key): correct reactant, correct product and correct balancing, 1 mark each (2023E-Q39(bi)); the conditions for hydration must be written (temperature 300 °C, pressure

6000–7000 kPa, H_2O as a gas, any one of these, 2022S-Q33(a)); the catalyst written as H^+ or H_3PO_4, H_2SO_4 (2021E-Q39(a)); colour change matched to the substance (2017S-Q35(b): bromine water decolourises rapidly with cis/trans alkenes, but not with cycloalkanes).

Common errors (examiner report)

- In hydration equations, water must be written as gaseous $H_2O(g)$ (the 2021E-Q39(a) key explicitly requires this).
- 2023E-Q37(d) (3 marks) asks students to identify the isomer that reacts with water to form a primary alcohol and write the equation. The key accepts but-1-ene (the C_5H_{10} version is pent-1-ene), with the equation $C_4H_8 + H_2O \rightarrow CH_3CH_2CH_2CH_2OH$. This answer differs from the general prediction from Markovnikov's rule: Markovnikov's rule states that when a 1-alkene is hydrated under H^+ catalysis, H adds to the terminal carbon and OH adds to the internal carbon, giving a secondary alcohol as the major product (butan-2-ol); to obtain a primary alcohol, anti-Markovnikov addition is needed (e.g. hydroboration followed by oxidation), but the question does not give that pathway. When answering, writing the primary alcohol as in the key earns the marks; however, do not memorise "hydration of a 1-alkene gives a primary alcohol" as a general rule. The 2023 report said Question 37 was generally answered well and did not comment separately on (d).
- Catalyst and conditions are marked separately: in 2021E-Q39(a), the key has a "state the catalyst" mark for each step, and 2022S-Q33(a) requires the conditions for hydration.

Past questions: 2016S-Q30(a), 2017S-Q26(bi), 2017S-Q26(bii), 2017S-Q35(b), 2019S-Q33, 2020S-Q33, 2021E-Q39(a), 2022S-Q33(a), 2023E-Q37(d), 2023E-Q39(bi)

Pattern 6: Oxidation of alcohols and aldehydes

Identify: The question gives an alcohol or aldehyde with acidified $KMnO_4$ or acidified $K_2Cr_2O_7$ (or $Na_2Cr_2O_7$), and asks for products, colour changes, equations or reaction type; or asks for the half-equation for oxidation.

Steps

1. Determine the class of alcohol: primary alcohol → aldehyde → carboxylic acid (the aldehyde requires further oxidation, excess oxidant, reflux); secondary alcohol → ketone (no further oxidation); tertiary alcohol → no reaction.
2. Aldehydes are easily oxidised to carboxylic acids (e.g. pentanal → pentanoic acid).
3. Colour changes: MnO_4^- (purple) is reduced to Mn^{2+} (almost colourless or pale pink); $Cr_2O_7^{2-}$ (orange) is reduced to Cr^{3+} (green).
4. Write half-equations: the oxidation half-equation of the organic compound (add H^+, H_2O, e^- and balance), plus the reduction half-equation of the oxidising agent, then combine into the overall reaction (see the balancing method in RX2).
5. State the reaction type: oxidation (the organic compound is oxidised) or redox.

Mark-earning points (typical wording in the key): correct products (2021S-Q27(a): 2-methylpentan-2-ol is no reaction, pentan-1-ol gives pentanal and pentanoic acid, pentan-2-ol gives pentan-2-one, 1 mark per row); complete description of the colour change (2016S-Q27(aiii): purple solution mixed with colourless alcohol, the purple colour fades; 2023E-Q37(eii): orange solution added gives a dark green solution; 2021S-Q27(cii) compares the colour changes of the two oxidising agents); species and balancing of the half-equations and overall reaction each score (2024S-Q33(b): $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$, $CH_3OH + H_2O \rightarrow HCOOH + 4H^+ + 4e^-$, overall reaction $5CH_3OH + 4MnO_4^- + 12H^+ \rightarrow 5HCOOH + 4Mn^{2+} + 11H_2O$); reaction type (2021S-Q27(ci): oxidation and reduction / redox; 2025E-Q38(ci): oxidation).

Common errors (examiner report)

- Tertiary alcohols do not react: 2021S-Q27(a) requires writing "no reaction", not any product.
- Distinguishing aldehydes from acids: an aldehyde can be oxidised further to an acid, so the conditions must be stated (excess oxidant); the 2021S-Q27(a) key gives pentanoic acid under the condition "excess".
- In 2024S-Q33(b), with excess MnO_4^- the product of methanol is written as methanoic acid (HCOOH),

not CO_2; when answering, follow the question and the key.

- The 2025E-Q38(cii) key accepts acidified potassium dichromate or acidified potassium permanganate as the reagent, and the acidic condition (H^+) must be written. The report did not comment separately on this question type.

Past questions: 2016S-Q27(aiii), 2016S-Q27(b), 2018S-Q30(c), 2019S-Q33, 2020S-Q33, 2021S-Q27(a), 2021S-Q27(b), 2021S-Q27(ci), 2021S-Q27(cii), 2022S-Q32(b), 2023E-Q37(ei), 2023E-Q37(eii), 2024S-Q33(b), 2025E-Q38(ci), 2025E-Q38(cii)

Pattern 7: Esterification and hydrolysis; reactions of carboxylic acids and amines

Identification: The question stem contains "esterification", "hydrolysis", "ester", "amide", "carboxylic acid with alcohol / carbonate / NaOH", or "amine"; or it requires deducing the alcohol or acid from the monomers or products.

Steps

1. Esterification: RCOOH + R′OH ⇌ RCOOR′ + H_2O, concentrated H_2SO_4 as catalyst, heat; write the equilibrium arrow ⇌.
2. Deducing starting materials from an ester: break the C(=O)–O bond, add OH to the acid end and H to the alcohol end.
3. Hydrolysis: an ester or amide bond is broken under acid or base conditions; the products of basic hydrolysis (saponification) are a carboxylate salt and an alcohol.
4. Acid–base reactions of carboxylic acids: RCOOH + OH^- → $RCOO^-$ + H_2O; reaction with carbonate releases CO_2.
5. Amines: $–NH_2$ is a base and can accept H^+ (forming $–NH_3^+$).

Scoring points (typical key wording): the esterification equation includes the structures of the alcohol, carboxylic acid and ester, equilibrium and the formation of water (2019E-Q38(b): 4 marks, must include water; 2020E-Q40(c): water listed as a product); for the reaction of sodium hydroxide with a carboxylic acid, write the carboxylate and water (2018E-Q40(b); 2020E-Q40(ei): soap and H_2O as products); when deducing an alcohol from an ester, state the name of the alcohol (2021E-Q34(e): ethanol); description of observations (2016S-Q27(ai): two colourless solutions mixed, producing a fruity smell).

Common errors (examiner reports)

- Esterification is reversible, so do not write only a single arrow; the key for 2020E-Q40(c) requires an equilibrium arrow.
- Omitting water. The key for 2019E-Q38(b) explicitly states the equation must include H_2O.
- When deducing an alcohol from an ester, omitting the ester's alkoxy group ($–OCH_2CH_3$ corresponds to ethanol), a common error in 2021E-Q34(e).
- 2016E-Q38(b) requires writing the two hydrolysis products of acetylcholine: a carboxylic acid (ethanoic acid) and choline (a cation). Both structures must be drawn.
- The report does not specifically name this pattern, but the key for 2023E-Q37(fi) requires 1 mark each for reactants and products, so these must be checked individually.

Past exam questions: 2016E-Q38(b), 2016S-Q27(ai), 2016S-Q30(a), 2016S-Q30(b), 2018E-Q40(b), 2019E-Q38(b), 2019S-Q33, 2020E-Q40(c), 2020E-Q40(ei), 2020S-Q33, 2021E-Q34(e), 2022S-Q33(b), 2023E-Q37(fi), 2025E-Q35(a)

Pattern 8: Designing a chemical test to distinguish organic compounds

Identification: The question stem asks you to "suggest a distinguishing test", "propose a chemical test", "state the reagent(s) and the observations", or fill in a table with reagents and observations.

Steps

1. Identify the functional group difference(s) between the two (or several) substances (C=C, primary

alcohol, aldehyde, carboxylic acid, cycloalkane, etc.).

2. Choose a reagent that gives a characteristic observation with one of the functional groups (e.g. Br_2(aq) with C=C; acidified potassium dichromate with a primary alcohol or aldehyde; sodium hydrogencarbonate with a carboxylic acid).
3. For each substance, write the expected observation (colour change, gas, precipitate).
4. In the table, the reagent, the observation for substance A and the observation for substance B — all three must be completed.

Scoring points (typical key wording): reagent correct (2017S-Q35(b): Br_2(aq) or another reasonable test); the observation for each substance is specific (cis/trans chain alkenes: orange solution rapidly decolourises; cyclic hydrocarbon: remains orange); 2018S-Q33(c) (butan-1-ol and butanoic acid) key requires 1 mark each for the reagent, the observation for the alcohol and the observation for the acid; 2022S-Q32(b) requires proposing a test and stating the observation for each of the two isomers.

Common errors (examiner reports)

- Writing only the reagent, not the specific observation, or writing only "reaction occurs" without describing colour, gas or precipitate.
- The chosen reagent cannot distinguish: e.g. using Br_2(aq) to distinguish two substances that are both saturated is ineffective.
- Note: the original key only gives "reagent + observation" scoring points; this note does not include the marking basis for other common tests (such as Tollens or Fehling reagent), so refer to the original paper.

Past exam questions: 2017S-Q35(b), 2018S-Q33(c), 2022S-Q32(b)

Pattern 9: Deducing structures from reactions and data

Identification: The question stem gives reaction observations, molecular formula or analytical data, and asks you to "use your calculated empirical formula to demonstrate" or deduce a product structure.

Steps

1. Write the empirical formula or molecular formula obtained from the data (the calculation part belongs to QNT5).
2. Write the molecular formulas of the expected product and the starting material.
3. Compare: if the calculated result matches the expected product but not the starting material, this shows the reaction proceeded as expected.
4. The conclusion must state what this comparison supports (e.g. the enzyme is still active).

Scoring points (typical key wording): compare the empirical formula with the expected product ($C_5H_{14}ON$, same as choline); the conclusion corresponds to the comparison, stating that it matches the product rather than the substrate, thereby proving the enzyme is active (2016E-Q38(d), 2 marks).

Common errors (examiner reports)

- Writing only "matches", without stating which substance it is compared with.
- The conclusion only states "enzyme is effective", without writing the contrast "matches the product, not the substrate".
- This pattern has only 1 part; the report has no specific comment, so there is limited evidence.

Past exam questions: 2016E-Q38(d)